

	GLOBAL ECONOMICS GRADE: 11TH TERM 3 — SOLVED MIDTERM HYPOTHESIS TEST DESIGN AND DECISION (C1–C7) TEACHER’S NAME: Nicolás López Cuéllar	THIRD TERM 2025–2026
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Learning objective: Design and apply hypothesis tests from real contexts by identifying the parameter, writing hypotheses, selecting the test direction, applying a proportion test and a mean test, interpreting statistical errors, and determining the minimum sample size needed to reject the null hypothesis.	Evaluated criteria: C1: Describes the concept of a statistical hypothesis. C2: Identifies the null and alternative hypotheses in a statistical test. C3: Determines whether a test should be one-tailed or two-tailed. C4: Applies a hypothesis test in a population proportion setting. C5: Applies a hypothesis test in a population mean counting setting. C6: Identifies Type I or Type II error and its consequences. C7: Finds the minimum sample size needed to reject H_0.
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Critical values used

$$z_{0.95} = 1.645 \quad z_{0.975} = 1.960 \quad z_{0.99} = 2.326$$

For right-tailed tests, reject when $z > z_{\text{critical}}$. For left-tailed tests, reject when $z < -z_{\text{critical}}$. For two-tailed tests, reject when $|z| > z_{\text{critical}}$.

Problem 1 — Grade 11 punctuality audit

The school compares punctuality between Prom 2025 and Prom 2026. The variable is the number of late arrivals to first class per morning.

1 = more than 9 late arrivals 0 = not more than 9 late arrivals

#	Prom 2025	2025-bin	Prom 2026	2026-bin
1	5	0	5	0
2	6	0	7	0
3	7	0	8	0
4	8	0	9	0
5	9	0	10	1
6	4	0	5	0
7	6	0	8	0
8	7	0	9	0
9	10	1	11	1
10	5	0	6	0
11	8	0	10	1
12	7	0	8	0
13	6	0	7	0
14	9	0	10	1
15	11	1	12	1

#	Prom 2025	2025-bin	Prom 2026	2026-bin
16	5	0	6	0
17	7	0	8	0
18	8	0	10	1
19	10	1	11	1
20	6	0	7	0
21	10	1	10	1
22	7	0	8	0
23	6	0	7	0
24	8	0	10	1
25	11	1	12	1
26	5	0	6	0
27	7	0	8	0
28	9	0	10	1
29	12	1	11	1
30	10	1	9	0

Summary data:

$$n_{25} = 30 \quad \bar{x}_{25} = 7.63 \quad \sigma_{25} = 2.09 \quad x_{25} = 7 \quad \hat{p}_{25} = \frac{7}{30} \approx 0.2333$$

$$n_{26} = 30 \quad \bar{x}_{26} = 8.60 \quad \sigma_{26} = 3.24 \quad x_{26} = 12 \quad \hat{p}_{26} = \frac{12}{30} = 0.40$$

Use $\alpha = 0.05$

C1

Mean setting

μ_{25} = true mean late arrivals for Prom 2025 μ_{26} = true mean late arrivals for Prom 2026

$$n_{25} = 30 \quad \bar{x}_{25} = 7.63 \quad \sigma_{25} = 3.40$$

$$n_{26} = 30 \quad \bar{x}_{26} = 8.60 \quad \sigma_{26} = 3.24$$

Proportion setting

p_{25} = true high-lateness proportion for Prom 2025 p_{26} = true high-lateness proportion for Prom 2026

$$x_{25} = 7 \quad n_{25} = 30 \quad \hat{p}_{25} = \frac{7}{30} \approx 0.2333$$

$$x_{26} = 12 \quad n_{26} = 30 \quad \hat{p}_{26} = \frac{12}{30} = 0.40$$

Using Prom 2025 as benchmark

C2

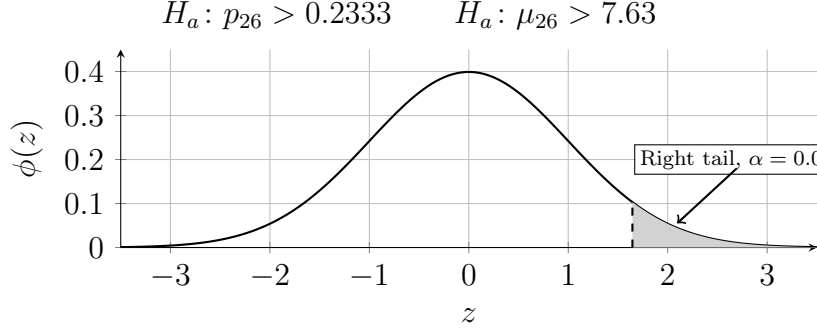
Benchmark direction: Prom 2026 vs Prom 2025

$$H_0: p_{26} = 0.2333 \quad H_a: p_{26} > 0.2333 \quad p_0 = 0.2333$$

$$H_0: \mu_{26} = 7.63 \quad H_a: \mu_{26} > 7.63 \quad \mu_0 = 7.63$$

C3

Right-tailed



C4

$$\hat{p}_{25} = 0.2333 \quad \hat{p}_{26} = 0.40 \quad n_{26} = 30 \quad k = 2.20$$

$$z = \frac{\hat{p}_{26} - \hat{p}_{25}}{\sqrt{k \frac{\hat{p}_{25}(1-\hat{p}_{25})}{n_{26}}}} = \frac{0.40 - 0.2333}{\sqrt{2.20 \frac{0.2333(0.7667)}{30}}} \approx 1.46$$

$$z_{\text{critical}} = 1.645 \quad 1.46 < 1.645$$

Fail to reject H_0 . $p_{26} > 0.2333$ is not supported.

C5

$$\bar{x}_{25} = 7.63 \quad \bar{x}_{26} = 8.60 \quad \sigma_{26} = 3.24 \quad n_{26} = 30$$

$$z = \frac{\bar{x}_{26} - \bar{x}_{25}}{\frac{\sigma_{26}}{\sqrt{n_{26}}}} = \frac{8.60 - 7.63}{\frac{3.24}{\sqrt{30}}} \approx 1.64$$

$$z_{\text{critical}} = 1.645 \quad 1.64 < 1.645$$

Fail to reject H_0 . $\mu_{26} > 7.63$ is not supported.

C7

Proportion test

$$\frac{0.1667}{\sqrt{2.20 \frac{0.2333(0.7667)}{n_{26}}}} > 1.645$$

$$n_{26} > \left(\frac{1.645 \sqrt{2.20 \cdot 0.2333(0.7667)}}{0.1667} \right)^2 \approx 38.34$$

$$n_{26, \min} = 39$$

Mean test

$$\frac{0.97}{\frac{3.24}{\sqrt{n_{26}}}} > 1.645 \Rightarrow n_{26} > \left(\frac{1.645(3.24)}{0.97} \right)^2 \approx 30.18$$

$$n_{26, \min} = 31$$

C6

Initial sample size $n_{26} = 30$: **Fail to reject H_0** . both tests do not support increases.

$$z_p = \frac{0.1667}{\sqrt{2.20 \frac{0.2333(0.7667)}{39}}} \approx 1.66 > 1.645$$

$$z_\mu = \frac{0.97}{\frac{3.24}{\sqrt{31}}} \approx 1.67 > 1.645$$

Minimum sample sizes: **Reject H_0** . both tests support increases. Type II error possible when failing to reject; Type I error possible when rejecting.

Using Prom 2026 as benchmark

C2

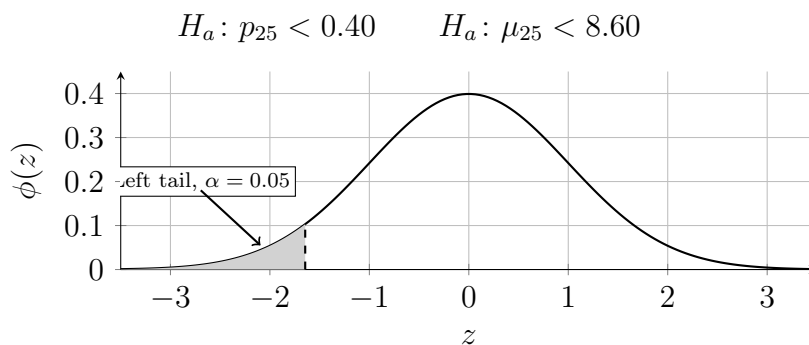
Benchmark direction: Prom 2025 vs Prom 2026

$$H_0: p_{25} = 0.40 \quad H_a: p_{25} < 0.40 \quad p_0 = 0.40$$

$$H_0: \mu_{25} = 8.60 \quad H_a: \mu_{25} < 8.60 \quad \mu_0 = 8.60$$

C3

Left-tailed



C4

$$\begin{aligned}\hat{p}_{26} &= 0.40 & \hat{p}_{25} &= 0.2333 & n_{25} &= 30 & k &= 1.50 \\ z &= \frac{\hat{p}_{25} - \hat{p}_{26}}{\sqrt{1.50 \frac{\hat{p}_{26}(1-\hat{p}_{26})}{n_{25}}}} = \frac{0.2333 - 0.40}{\sqrt{1.50 \frac{0.40(0.60)}{30}}} \approx -1.52 \\ z_{\text{critical}} &= -1.645 & -1.52 &> -1.645\end{aligned}$$

Fail to reject H_0 . $p_{25} < 0.40$ is not supported.

C5

$$\begin{aligned}\bar{x}_{26} &= 8.60 & \bar{x}_{25} &= 7.63 & \sigma_{25} &= 3.40 & n_{25} &= 30 \\ z &= \frac{\bar{x}_{25} - \bar{x}_{26}}{\frac{\sigma_{25}}{\sqrt{n_{25}}}} = \frac{7.63 - 8.60}{\frac{3.40}{\sqrt{30}}} \approx -1.56 \\ z_{\text{critical}} &= -1.645 & -1.56 &> -1.645\end{aligned}$$

Fail to reject H_0 . $\mu_{25} < 8.60$ is not supported.

C7

Proportion test

$$\frac{0.1667}{\sqrt{1.50 \frac{0.40(0.60)}{n_{25}}}} > 1.645 \Rightarrow n_{25} > \left(\frac{1.645 \sqrt{1.50 \cdot 0.40(0.60)}}{0.1667} \right)^2 \approx 44.54$$

$$n_{25, \min} = 45$$

Mean test

$$\frac{0.97}{\frac{3.40}{\sqrt{n_{25}}}} > 1.645 \Rightarrow n_{25} > \left(\frac{1.645(3.40)}{0.97} \right)^2 \approx 33.23$$

$$n_{25, \min} = 34$$

C6

Initial sample size $n_{25} = 30$: **Fail to reject H_0 .** both tests do not support decreases.

$$z_p = \frac{-0.1667}{\sqrt{1.50 \frac{0.40(0.60)}{45}}} \approx -1.66 < -1.645$$

$$z_\mu = \frac{-0.97}{\frac{3.40}{\sqrt{34}}} \approx -1.66 < -1.645$$

Minimum sample sizes: **Reject H_0 .** both tests support decreases. Type II error possible when failing to reject; Type I error possible when rejecting.